

Aditya Chauhan

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EDUCATION

Dr. B.R. Ambedkar National Institute of Technology, Jalandhar
Bachelor of Technology in Computer Science and Engineering

Aug 2023 – Jun 2027 (Expected)
CGPA: 8.53/10

TECHNICAL SKILLS

- **Languages:** C++, Python, C, JavaScript
- **Frameworks:** React.js, Node.js, Express.js, FastAPI, TailwindCSS, DaisyUI
- **Databases:** MongoDB, MySQL, ChromaDB
- **AI/ML:** RAG, LangChain, Langfuse, OpenTelemetry, Scikit-learn, Pandas, SHAP, SMOTE
- **Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, System Design
- **Tools:** Git, GitHub, Postman, REST APIs, WebSockets, JWT

EXPERIENCE

Accenture Jun 2026 – Jul 2026
Software Development Engineer Intern Bengaluru

- Decreased RAG input-token usage by **34%** (1800 → 1180 tokens/request) for **ContextIQ** using LangChain chunking, validated with Langfuse.
- Slashed vector-search latency by **69%** (580ms to 180ms) by tracing a **400ms** bottleneck with OpenTelemetry and refactoring the ChromaDB client into a singleton.

Google Developer Group Dec 2023 – Present
Software Development & CP Core Team Co-Lead NIT Jalandhar

- Increased contest performance by **40%** and junior participation by **30%** by mentoring junior members through structured DSA sessions and contest reviews.
- Coordinated **CodeVerse** and **CodeHunt** for **90+** participants, curating **15** algorithmic problems while streamlining contest execution.

PROJECTS

SwiftCache – Thread-Safe Concurrent LRU Cache Server May 2026 – Jun 2026
C++, TCP/HTTP, Concurrency [GitHub](#)

- Constructed a thread-safe TCP-based LRU cache supporting deterministic **O(1)** operations using custom hash-map, linked-list and TTL eviction.
- Sustained **15,000+ QPS** with **<3ms** p99 latency through benchmark-driven optimization of concurrent read/write workloads.
- Designed a modular caching architecture with self-serve configuration, reducing feature integration effort by **3x**.

ChatMind – Real-Time Messaging Platform with RAG Q&A Mar 2026 – May 2026
MERN, FastAPI, WebSockets [Live Demo](#) | [GitHub](#)

- Accelerated end-to-end messaging latency by **20%** to **<50ms** using WebSockets for real-time bidirectional communication.
- Architected a hybrid RAG pipeline using BM25, vector retrieval and RRF for contextual question answering.
- Boosted context relevance scores (MRR@5) by **35%** by deploying cross-encoder reranking for more accurate retrieval results.

RetainOps – AI-Powered Customer Retention Platform Jan 2026 – Feb 2026
React, FastAPI, Scikit-learn, SHAP [Live Demo](#) | [GitHub](#)

- Developed a full-stack customer retention platform integrating React and FastAPI, deploying production-ready ML inference.
- Identified **63.8%** of revenue-at-risk while targeting the top **20%** of customers using a custom Master Model Score across **5** tuned classifiers.
- Improved prediction reliability by engineering **9** features from **7,100+** records while mitigating a **3:1** class imbalance using SMOTE and SHAP explainability.

COMPETITIVE PROGRAMMING & ACHIEVEMENTS

Profiles: [Codeforces](#) | [LeetCode](#) | [GeeksforGeeks](#) | [CodeChef](#)

- Qualified for **Meta Hacker Cup Round 2**, securing Global Rank **3223** (AIR **875**) in Round 1.
- Reached **Knight** on LeetCode (Max Rating: **1957**), achieving Global Rank **315** (Top **1.5%**) in Biweekly Contest 172.
- Attained **Specialist** on Codeforces (Max Rating: **1418**) and **3-Star** on CodeChef (Max Rating: **1661**).
- Ranked as a **Flipkart Grid 7.0 Semi-Finalist**, placing in the top **0.5%** among **1.5 lakh+** participants.